SAROJ ADHIKARI
Department of Physics
State University of New York at Plattsburgh, Plattsburgh, NY
Phone: (518) 564-2644
Email: saroj.adhikari@plattsburgh.edu

Employment

- **Assistant Professor of Physics** **2020 - present**
  Department of Physics, State University of New York at Plattsburgh
  Plattsburgh, New York
- **Visiting Assistant Professor of Physics** **2019 - 2020**
  Department of Electrical Engineering and Physics, Wilkes University
  Wilkes-Barre, Pennsylvania
- **Postdoctoral Research Fellow in Theoretical Cosmology** **2016 - 2019**
  Department of Physics and Leinweber Center for Theoretical Physics
  University of Michigan, Ann Arbor, Michigan

Education

- **PhD in Physics** **2010-2016**
  Pennsylvania State University, University Park, Pennsylvania
  Thesis: "Effects of primordial non-Gaussianity in the CMB and the large-scale structure"
- **BS in Physics** **2006-2010**
  Minors in Mathematics and Honors Interdisciplinary Studies
  University of Central Arkansas, Conway, Arkansas

Teaching

1. SUNY Plattsburgh
   - PHY 112 - General Physics II, Spring 2021
   - PHY 421 - Fundamentals of Electromagnetism, Spring 2021
   - CSC 322 - Data Modeling and Analysis, Fall 2020
2. Wilkes University
   - PHY 201 - General Physics I, Spring 2020
   - PHY 203 - Modern Physics, Fall 2019 and Spring 2020
   - PHY 311 - Thermodynamics & Statistical Mechanics, Fall 2019
3. Other Teaching
   - Guest Lecturer for undergraduate courses - Particles and Cosmology, Spring 2018 (Michigan) and Theoretical Mechanics, Spring 2014 (Penn State)
   - Recitation and Lab Teaching Assistant at Penn State for Introductory physics courses, Fall 2010, Spring 2011, Fall 2012, Spring 2013, Summer 2013, Spring 2016
   - Pedagogical Associate, Honors College Writing Seminar, Fall 2009, University of Central Arkansas

Research Interests

- **Early-universe physics**: primordial inflation and non-Gaussianity
- **Cosmological data analysis**: cosmic microwave background and large-scale structure
- **Statistical methods**: measures to quantify discrepancies between experiments

Publications

1. **S. Adhikari** and D. Huterer, "Super-CMB fluctuations and the Hubble tension," Phys. Dark. Univ. 28 (2020) 100539, arXiv:1905.02278
2. **S. Adhikari** and D. Huterer, "A new measure of tension between experiments," JCAP 01 (2019) 036, arXiv:1806.04292
3. X. Li, N. Weaverdyck, **S. Adhikari**, D. Huterer, J. Muir and H-Y. Wu, "The Quest for the Inflationary Spectral Runnings in the Presence of Systematic Errors," ApJ, 862 137 (2018), arXiv:1806.02515
4. J. Muir, **S. Adhikari** and D. Huterer, "Covariance of CMB anomalies," Phys. Rev. D98, 023521 (2018), arXiv:1806.02354
5. **S. Adhikari**, A.-S. Deutsch and S. Shandera, "Statistical anisotropies in the temperature and polarization fluctuations from a scale-dependent trispectrum," Phys. Rev. D98, 023520 (2018), arXiv:1805.00037
6. **S. Adhikari**, D. Jeong and S. Shandera, "Constraining primordial and gravitational mode coupling with the position-dependent bispectrum of the large-scale structure," Phys. Rev. D94, 083528 (2016), arXiv:1608.05139
7. **S. Adhikari**, S. Shandera and A.L. Erickcek, "Large-scale anomalies in the CMB as signatures of non-Gaussianity," Phys. Rev. D93, 023524 (2016), arXiv:1508.06489
8. **S. Adhikari**, "Local variance asymmetries in Planck temperature anisotropy map," MNRAS 446 (4), 4232-4338 (2015), arXiv:1408.5396
9. A.B. Mantz et al., "Weighing the Giants IV: Cosmology and Neutrino Mass," MNRAS 446 (3), 2205-2225 (2014), arXiv:1407.4516
10. **S. Adhikari**, S. Shandera and N. Dalal, "Higher moments of primordial non-Gaussianity and N-body simulations," JCAP 06 (2014) 052, arXiv:1402.2336

Online publication database links

- **ORCID**: https://orcid.org/0000-0002-3814-7708
- **INSPIRE**: https://inspirehep.net/author/profile/Saroj.Adhikari.1
- **Google scholar**: https://scholar.google.com/citations?user=AIpOzQ0AAAAJ

Computing Skills

- Proficient in Python, Mathematica, LaTeX
- Good knowledge of C, MPI and parallel computing on large clusters
- Experience with numerical optimization and scientific programming
- Experience with GADGET2 (N-body simulations) and HEALPIX (2D pixelization)
- Experience with Fisher forecasts and Markov-Chain Monte Carlo (MCMC) techniques

Mentoring

- **Mentor**, Undergraduate Research Opportunity Program (UROP)
  University of Michigan, Ann Arbor, Michigan 2018-2019

Awards

1. Google Cloud Research Credit, estimated value: $5,000.00 2020-2021
2. XSEDE Allocation Renewal, estimated value: $1,347.36 2018-2019
3. XSEDE Startup allocation, 100,000 cpu-hours, estimated value: $1,841.00 2017-2018
4. Department of Physics Awards and Scholarships, Penn State University
   - Frymoyer Fellowship in Physics 2015 – 2016
   - Downsborugh Graduate Fellowship in Physics 2014 – 2015
   - David C. Duncan Graduate Fellowship in Physics 2013 – 2014
   - David H. Rank Memorial Physics Award 2011 – 2012
   - Homer F. Braddock Scholarship 2010 – 2011
5. Research Experience for Undergraduates (REU)
   Kansas State University, Manhattan, Kansas 2009
6. Research Experience for Undergraduates (REU)
   Arkansas Center for Space and Planetary Sciences, Fayetteville, Arkansas
7. Google Summer of Code (declined) 2008
8. Dorothy M. Long Scholarship
   University of Central Arkansas Alumni Association, Conway, Arkansas 2008-2009
9. Honors College Full Tuition Award
   University of Central Arkansas Honors College, Conway, Arkansas 2006-2010

Research Talks

1. “Scale-dependent primordial trispectrum and cosmological parameter constraints"
   COSMO-18, Institute for Basic Science (IBS), Daejeon, Korea Aug 2018
2. “Primordial trispectrum and modulations in the CMB"
   TeVPA, Ohio State University, Columbus, Ohio Aug 2017
3. “Position-dependent bispectrum of the large-scale structure"
   Neighborhood Workshop in Cosmology, Penn State University Apr 2016
4. “Non-Gaussianity and statistical anisotropy"
   Rust-belt cosmology and high-energy physics meeting, Buffalo, New York Nov 2015
5. “Cosmic microwave background power asymmetry and non-Gaussianity"
   Cosmology seminar, University of Michigan, Ann Arbor, MI Oct 2015
6. “Cosmic microwave background power asymmetry and non-Gaussianity"
   Cosmology seminar, Perimeter Institute, Waterloo, Canada Oct 2015
7. “Power asymmetry from mode coupling"
   Neighborhood Workshop in Cosmology, Penn State University Mar 2015
8. “Higher moments of primordial non-Gaussianity"
   Neighborhood Workshop in Cosmology, Penn State University Apr 2015

9. “Cosmological simulations with feeder-type non-Gaussian initial conditions"
   Neighborhood Workshop in Cosmology, Penn State University — Feb 2013

Popular Science Articles and Talks

1. “Black holes in our galaxy and the Universe"
   Association of Nepali Physicists in America (ANPA) Newsletter — Jan 2021
2. “Cosmological Inflation and the Big Bang Puzzles"
   Saturday Morning Physics, University of Michigan, Ann Arbor, Michigan — Nov 2016

Posters

1. “Higher moments of primordial non-Gaussianity"
   Meeting of the Mid-Atlantic Section of the APS, University Park, Pennsylvania — Oct 2014
2. “Effect of primordial non-Gaussianity in the number counts"
   International HPC Summer School, Budapest, Hungary — Jun 2014

Schools and Workshops

1. “Innovative Cosmological Simulations with Machine Learning and Statistics"
   Carnegie Mellon University, Pittsburgh, Pennsylvania — Jun 2015
2. “Workshop on CMB large-angle anomalies"
   Case Western Reserve University, Cleveland, Ohio — Sep 2014
3. “International High Performance Computing Challenges Summer School"
   National Information Infrastructure Development Institute, Budapest, Hungary — Jun 2014
4. “Essential Cosmology for the Next Generation"
   Cosmology on the Beach, Los Cabos, Mexico — Jan 2014
5. “Workshop on Cosmic Acceleration"
   Carnegie Mellon University, Pittsburgh, Pennsylvania — Aug 2012